NGUYEN VAN HUAN

COMPUTER ENGINEERING STUDENT | COMPUTER VISION & INTELLIGENT TRANSPORTATION

Hanoi, Vietnam | +84 964 483 919 | huan.elenoa@gmail.com

GitHub | LinkedIn | Portfolio

PROFILE

Computer Engineering student at HUST (GPA 3.94/4.00) focused on computer vision and perception-to-control systems for intelligent transportation. First author of a manuscript currently under review at the International Journal of Intelligent Transportation Systems Research, with hands-on experience in Python, PyTorch, YOLO11n, OpenCV, and 3-D Kalman filtering. Interested in Computer Vision, Machine Learning, and Research Engineering internships.

EDUCATION

Hanoi University of Science and Technology (HUST) — **Hanoi, Vietnam | 2024–2028**

B.Eng. in Computer Engineering | GPA: 3.94/4.00 | TOEIC: 730

RESEARCH EXPERIENCE

Research Team Lead & First Author — AuraBeam, HUST Research Group — **Hanoi, Vietnam | 04/2026–05/2026**

- Led a small team in building and evaluating an adaptive driving beam research prototype.
- Developed a Python perception-to-control pipeline combining primary YOLO11n detection, a temporal depth prior, adaptive six-state 3-D Kalman filtering, and Arduino-compatible 8 × 8 LED hardware-in-the-loop evaluation.
- Improved D7 M-GSSR from 26.10% to 44.49% and reduced D6 false-darkening rate from 64.82% to 44.56% through control-target filtering.
- Benchmarked CPU inference and documented the latency bottleneck that must be resolved before 30-FPS deployment.

Manuscript: *"Robust Nighttime Vehicle Localization Under Headlight Saturation via YOLO Detection and Actuation-Aware 3-D Tracking."* Under review at the International Journal of Intelligent Transportation Systems Research.

SELECTED PROJECTS

Media Crisis Sentiment Analysis | *Python, PyTorch, PhoBERT, SVM, scikit-learn* — **03/2026**

- Built and evaluated PhoBERT/SVM classifiers on approximately 20,000 Vietnamese sentences using an 80/10/10 train-validation-test split; achieved an F1-score of 0.83 and contributed experimental results to a research manuscript.

Windows Tiny Shell | *C, Windows APIs, Process Management, CLI* — **12/2025**

- Developed a Windows command-line shell from a clearly attributed forked C codebase, covering process launching, built-in commands, execution-state monitoring, and background-task termination through native Windows APIs.

Dating Platform | *MongoDB, Express, React, Node.js* — **10/2025**

- Built and deployed a MERN web application with REST APIs, a MongoDB data model, a Render backend, and a React frontend hosted on Vercel.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript

Computer Vision & ML: PyTorch, YOLO11n, OpenCV, scikit-learn, NumPy, Pandas, PhoBERT, SVM

Systems & Tools: 3-D Kalman filtering, Arduino-compatible HIL, Windows APIs, Docker, Git, WSL, VMs

Web & Deployment: MongoDB, Express, React, Node.js, Vercel, Render